
ARTIFICIAL LIGHT AT NIGHT AND BAT ACTIVITY
ALONG A DISTANCE GRADIENT FROM THE COLTER BAY PARKING LOT
Grand Teton National Park, 2022
Nigel K. Anderson · American Museum of Natural History

Contents

1 Overview	2
2 Primary Model	2
3 Post-hoc: Color × Intensity	4
4 Post-hoc: Intensity within Color	6
5 Post-hoc: Intensity × Distance	6
6 Community Analysis	8
7 Species-level Results	10
8 Bat Morphology Traits	11
9 Moth (Lepidoptera) Data	11
9.1 Data and General Activity Model	11
9.2 Family-Level Models	13
9.3 Community Analysis	13
10 Bat vs. Moth Comparison	14
11 Summary	17
12 Species Key	17
Supplemental Materials	19

OVERVIEW

Study question: does light color (red vs. white) affect bat activity, and does the answer depend on intensity or distance from the Colter Bay parking lot?

Design: 14 acoustic monitoring stations along a distance gradient (<0.52 km to >2.1 km from Colter Bay parking lot), red vs. white light crossed at five intensities (10, 30, 50, 70, 100%). Survey season: 2022. Six focal species: Epfu, Laci, Lano, Myev, Mylu, Myvo.

Six species with sufficient data are further characterized by four continuous wing-morphology traits (ear height, forearm length, wing loading, aspect ratio). Second, the same light treatments were run past moth (Lepidoptera) data at all 14 stations over the same season, identified to species, allowing a direct bat-vs-moth comparison of the light color, intensity, and distance effects (Sections 9-10).

Additional supporting figures are collected in Supplemental Materials at the end.

Overall main results: Bats were most active under red light, but only when it was dim to moderately bright. At the highest brightness, red and white lighting made no difference to the bats. Bats were more active further from the bright Colter Bay parking lot, but this pattern faded out under high brightness. The smaller Myotis species were the ones that showed up further away from the parking lot while the larger bats were more prevalent at Colter Bay.

Moths showed the opposite color preference. They preferred white light at high light intensity. However, we did find that moths were less prevalent the closer the sites were to Colter Bay.

We of course have more results but I think these are the main findings. We find that moon phase and seasonality impact bat detections (differently for red and white). We also find that some species are being impacted by sky brightness at the different sites and also the moon phase.

PRIMARY MODEL

Family: nbinom2 (log)
Formula: detections ~ color * intensity + intensity * dist_km +
mean_phase + pct_nonforest + brightness_dark +

```

jd_c + I(jd_c^2) + (1 | site)
Zero infl: ~1
Data:      data_env

```

AIC	BIC	logLik	-2*log(L)	df.resid
24256.7	24398.1	-12105.4	24210.7	3434

Random effects:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	0.3728	0.6106

Number of obs: 3457, groups: site, 14

Dispersion parameter for nbinom2 family (): 0.607

Conditional model:

	Estimate	Std.Err	z value	Pr(> z)	
(Intercept)	2.1920095	0.3045	7.199	6.05e-13	***
colorW	0.0499199	0.0931	0.536	0.5917	
intensity30	0.7155529	0.1476	4.849	1.24e-06	***
intensity50	0.4606876	0.1569	2.936	0.0033	**
intensity70	0.6229863	0.1099	5.671	1.42e-08	***
intensity100	0.2331022	0.1331	1.752	0.0798	.
dist_km	0.1375974	0.0675	2.040	0.0414	*
mean_phase	0.1992094	0.1171	1.701	0.0889	.
pct_nonforest	0.5326479	0.8559	0.622	0.5337	
brightness_dark	-0.0262336	0.0376	-0.698	0.4854	
jd_c	-0.0073183	0.0017	-4.287	1.81e-05	***
I(jd_c^2)	-0.0003733	0.0002	-2.150	0.0315	*
colorW:intensity30	-0.7703147	0.1751	-4.399	1.09e-05	***
colorW:intensity50	-0.5357506	0.1459	-3.672	0.0002	***
colorW:intensity70	-0.1550017	0.1366	-1.135	0.2565	
colorW:intensity100	0.0172641	0.1353	0.128	0.8985	
intensity30:dist_km	-0.0501091	0.0319	-1.572	0.1159	
intensity50:dist_km	-0.0364428	0.0294	-1.239	0.2153	
intensity70:dist_km	-0.1275481	0.0272	-4.687	2.77e-06	***
intensity100:dist_km	-0.0897028	0.0269	-3.335	0.0009	***

Zero-inflation model:

Estimate	Std.Err	z value	Pr(> z)
----------	---------	---------	----------

(Intercept) -21.53 805.37 -0.027 0.979

Reference levels: color = R (red), intensity = 10%. Distance entered as continuous raw km (dist_km). Two 2-way interactions: color × intensity, intensity × dist_km.

POST-HOC: COLOR X INTENSITY

Estimated marginal means on the response scale (detections/night), averaged over distance. Holm-corrected pairwise contrasts.

=== Estimated marginal means: color at each intensity (response scale) ===

intensity = 10:

color	response	SE	df	asympt.LCL	asympt.UCL
R	12.7	2.29	Inf	8.93	18.1
W	13.4	2.42	Inf	9.37	19.0

intensity = 30:

color	response	SE	df	asympt.LCL	asympt.UCL
R	23.2	4.64	Inf	15.70	34.3
W	11.3	2.31	Inf	7.57	16.9

intensity = 50:

color	response	SE	df	asympt.LCL	asympt.UCL
R	18.6	3.90	Inf	12.30	28.0
W	11.4	2.45	Inf	7.49	17.4

intensity = 70:

color	response	SE	df	asympt.LCL	asympt.UCL
R	17.8	3.21	Inf	12.49	25.4
W	16.0	2.97	Inf	11.14	23.0

intensity = 100:

color	response	SE	df	asympt.LCL	asympt.UCL
R	13.1	2.48	Inf	9.05	19.0
W	14.0	2.51	Inf	9.87	19.9

=== Red vs. White contrast at each intensity (Holm-corrected) ===

intensity	ratio	SE	df	z.ratio	p.value
10	0.951	0.089	Inf	-0.536	0.5917
30	2.055	0.299	Inf	4.946	<0.0001 ***
50	1.626	0.182	Inf	4.345	<0.0001 ***
70	1.111	0.109	Inf	1.067	0.2860
100	0.935	0.095	Inf	-0.665	0.5062

Ratio = R / W (ratio > 1 = more detections under red)

At 30% intensity, red light produced 2.06x more detections than white ($p < 0.0001$). At 50%, 1.63x ($p < 0.0001$). At 70% and 100%, no significant difference. White and red converge above 50% intensity (Supplemental Figs. S1-S2).

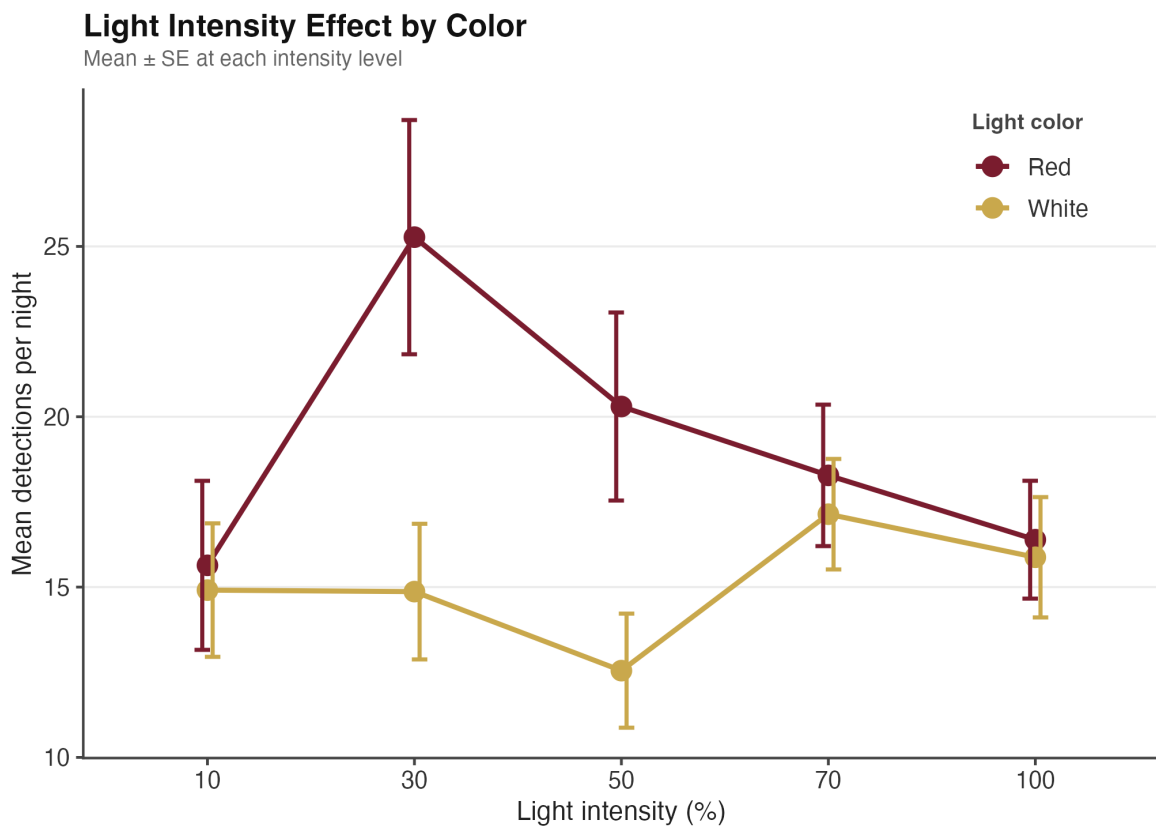


Figure 1: Color $\times$ intensity interaction: predicted detections from the GLMM on the response scale. Cardinal crimson = red light; gilded gold = white light.

POST-HOC: INTENSITY WITHIN COLOR

=== Intensity pairwise contrasts within Red (Holm-corrected) ===

contrast	ratio	SE	df	z.ratio	p.value	
10 / 30	0.547	0.075	Inf	-4.406	0.0001	***
10 / 50	0.685	0.098	Inf	-2.654	0.0478	*
10 / 70	0.714	0.066	Inf	-3.619	0.0024	**
10 / 100	0.969	0.113	Inf	-0.272	1.0000	
30 / 50	1.251	0.192	Inf	1.457	0.4350	
30 / 70	1.305	0.182	Inf	1.912	0.2237	
30 / 100	1.771	0.231	Inf	4.387	0.0001	***
50 / 70	1.043	0.148	Inf	0.298	1.0000	
50 / 100	1.415	0.171	Inf	2.880	0.0278	*
70 / 100	1.356	0.163	Inf	2.539	0.0556	.

=== Intensity pairwise contrasts within White (Holm-corrected) ===

contrast	ratio	SE	df	z.ratio	p.value
10 / 30	1.182	0.177	Inf	1.119	1.0000
10 / 50	1.170	0.184	Inf	0.999	1.0000
10 / 70	0.834	0.085	Inf	-1.778	0.6033
10 / 100	0.952	0.096	Inf	-0.484	1.0000

[all remaining contrasts: p >= 0.14]

Under red light, activity peaks at 30% (10/30 ratio = 0.547, i.e. 30% produces 1.83x more than 10%; $p < 0.0001$) and collapses at 100% (30/100 ratio = 1.77; $p < 0.0001$). 50% and 70% are intermediate and not distinguishable from 30%. White is flat across all intensities -- no pairwise contrast survives Holm correction (Supplemental Figs. S3-S4).

POST-HOC: INTENSITY $\times$ DISTANCE

Distance entered as continuous raw km. Interaction tests whether the effect of intensity on detections changes with distance from the Colter Bay parking lot.

=== Distance slope (km) by intensity level (emtrends) ===

intensity	dist_km.trend	SE	df	asympt.LCL	asympt.UCL	
10	0.1376	0.0675	Inf	0.005	0.270	*
30	0.0875	0.0689	Inf	-0.047	0.222	
50	0.1012	0.0676	Inf	-0.031	0.234	
70	0.0100	0.0670	Inf	-0.121	0.141	
100	0.0479	0.0669	Inf	-0.083	0.179	

Slopes averaged over color levels. Positive = more detections farther from Colter Bay parking lot (recovery with distance). Only intensity 10 slope differs from zero (95% CI excludes 0).

=== Pairwise distance-slope contrasts (Holm-corrected) ===

contrast	estimate	SE	df	z.ratio	p.value	
intensity10 - 70	0.1275	0.027	Inf	4.687	<0.0001	***
intensity10 - 100	0.0897	0.027	Inf	3.335	0.0077	**
intensity50 - 70	0.0911	0.028	Inf	3.222	0.0102	*

[all other contrasts: p >= 0.085]

=== Intensity contrasts at ecologically meaningful distances ===

dist_km = 0.3 km (Close):

contrast	ratio	SE	df	z.ratio	p.value	
intensity10 / 70	0.602	0.051	Inf	-6.043	<0.0001	***
intensity70 / 100	1.340	0.133	Inf	2.944	0.0292	*

[all other contrasts: p >= 0.10]

dist_km = 0.7 km (Medium):

contrast	ratio	SE	df	z.ratio	p.value	
intensity10 / 70	0.634	0.050	Inf	-5.822	<0.0001	***
intensity70 / 100	1.320	0.125	Inf	2.938	0.0297	*

[all other contrasts: p >= 0.13]

dist_km = 2.0 km (Far):

contrast	ratio	SE	df	z.ratio	p.value	
intensity10 / 70	0.748	0.052	Inf	-4.176	0.0003	***

[all other contrasts: p >= 0.07]

The intensity $\times$ distance interaction is driven by 70% and 100%: their distance slopes are significantly shallower than 10% (i.e. high intensities suppress bats uniformly across distance, not just close in). At 0.3 km, 70% produces 1.66x more detections than 10% (ratio = 0.602 inverted); this gap narrows but persists at 2.0 km (ratio = 0.748, $p = 0.0003$). 70% and 100% converge only at far distances (additional distance-gradient views in Supplemental Figs. S5–S7).

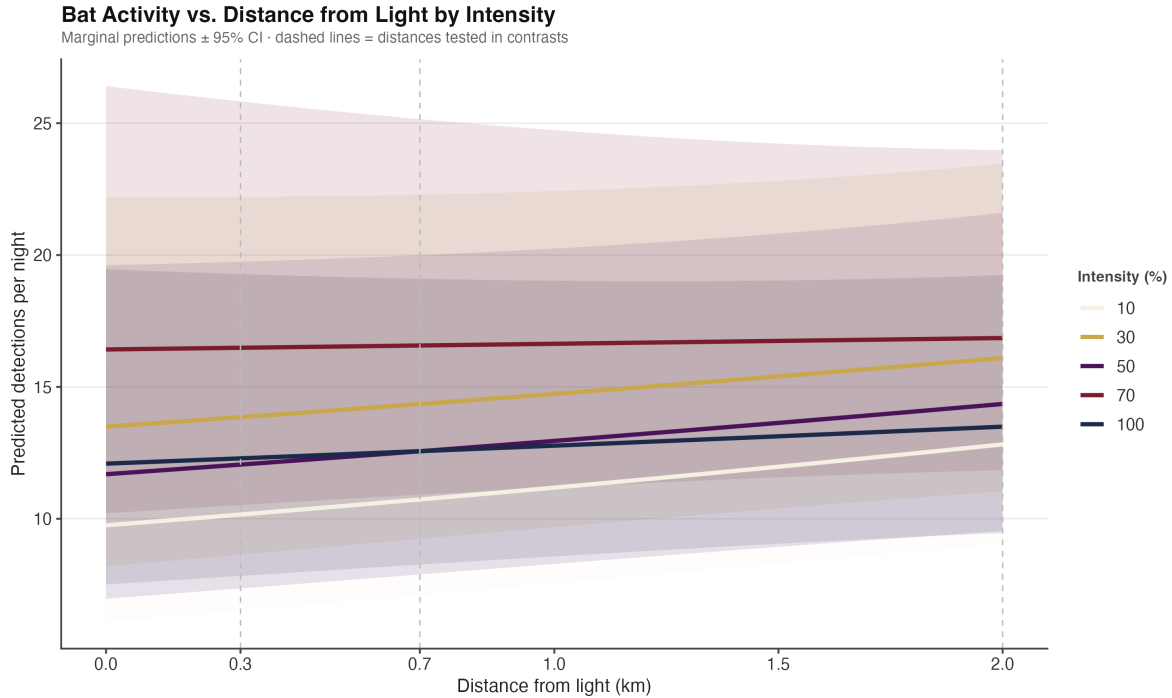


Figure 2: Bat activity vs. distance from light by intensity. Marginal predictions $\pm$ 95% CI; dashed vertical lines mark the 0.3, 0.7, and 2.0 km distances tested in the pairwise contrasts above.

COMMUNITY ANALYSIS

=== NMDS ===

Stress: 0.207 (k = 2, Bray-Curtis, trymax = 100)

=== PERMANOVA (adonis2) ===

	Df	SumOfSqs	R2	F	Pr(>F)
Model	9	3.8338	0.156	2.507	<0.001 ***
Residual	122	20.7295	0.844		
Total	131	24.5633	1.000		


```

=== Homogeneity of dispersion (by color) ===
      Df    Sum Sq   Mean Sq  F value  Pr(>F)
Groups    1  0.00014  0.000145   0.0061  0.938
Residuals 130  3.07863  0.023682

```

```

=== dbRDA marginal tests ===
      Df  SumOfSqs      F    Pr(>F)
color      1    0.2421  1.425    0.193
intensity   4    1.1300  1.663    0.048  *
dist_km     1    1.2271  7.222 <0.001  ***
mean_phase  1    0.2509  1.477    0.184
pct_nonforest 1    0.6241  3.673    0.007  **
brightness_dark 1    0.5661  3.332    0.012  *
Residual   122   20.7295

```

```

=== mvabund -- multivariate GLM (nBoot = 9999) ===
      Dev   Df  Pr(>Dev)
color    9.22   1   0.415
intensity 83.52   4   0.002  **
dist_km  142.57   1 <2e-16  ***
mean_phase 24.59   1   0.016  *
pct_nonforest 40.16   1   0.001  ***
brightness_d 54.29   1 <2e-16  ***

```

Color does not restructure community composition (dbRDA $p = 0.193$, mvabund $p = 0.415$). Distance is the dominant structuring variable (dbRDA $F = 7.22$, $p < 0.001$; mvabund Dev = 142.6, $p < 0.001$). Intensity marginally significant in both analyses. Habitat and sky brightness also contribute. Dispersion homogeneous between color treatments ($p = 0.938$). PERMANOVA result reflects centroid location, not spread.

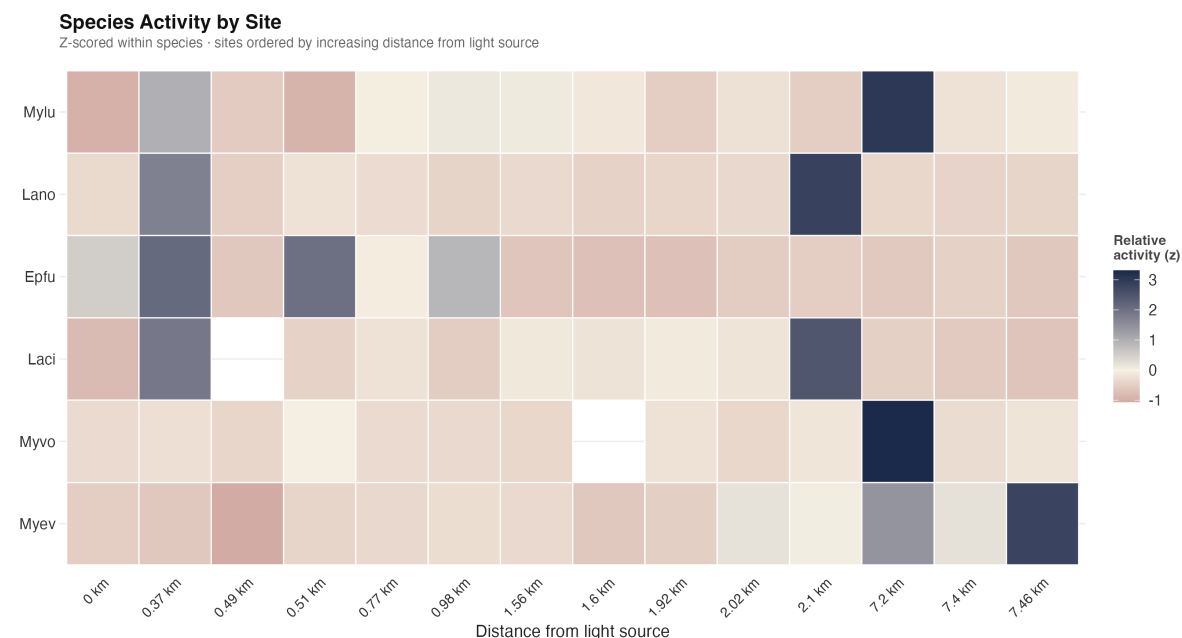


Figure 3: Species activity by site (z-scored within species), sites ordered by increasing distance from the Colter Bay parking lot.

SPECIES-LEVEL RESULTS

=== mvabund univariate p-values (adjusted, PIT-trap 9999 iter) ===

Species	Color	Intensity	Distance	Moon phase	%nonforest	brightness
Epfu	ns	***	**	*	ns	ns
Laci	ns	ns	**	ns	ns	ns
Lano	ns	ns	**	ns	ns	ns
Myev	ns	ns	***	ns	**	ns
Mylu	ns	ns	*	ns	ns	ns
Myvo	ns	ns	***	ns	*	***
Myyu	ns	ns	ns	ns	ns	ns
Myci	ns	ns	ns	ns	ns	ns

Epfu: most treatment-sensitive; responds to intensity, distance, and moon phase. Myev: tracks habitat (pct_nonforest) and distance. Myvo: strongly tracks sky brightness and distance. Color is non-significant for all species (Supplemental Figs. S8-S12).

BAT MORPHOLOGY TRAITS

We have four continuous wing-morphology measurements per species: ear height, forearm length, wing loading, and aspect ratio. Each trait was modeled separately (z-scored, interacted with intensity and distance) and compared at representative low / mean / high (± 1 SD) values.

All four traits point the same direction. Species with smaller ears, shorter forearms, lower wing loading, and lower aspect ratio -- slower, more maneuverable fliers such as *Myotis* spp. -- show an increase in activity with distance from the Colter Bay parking lot, while species at the high end of each trait (larger, faster, open-air-adapted fliers such as *Lasi*) are comparatively flat across the same gradient. The same ordering holds for intensity: low-trait-value species have consistently higher predicted detections at every intensity level, with the gap narrowing at high intensity. Because forearm length, wing loading, ear height, and aspect ratio are all measured on the same six species, the four traits are highly correlated with one another (and with species identity itself, as guild was) -- the four panels are best read as one wing-morphology gradient viewed four ways, not four independent effects (Supplemental Figs. S13-S14).

MOTH (LEPIDOPTERA) DATA

Data and General Activity Model

Moth specimens were identified from nightly traps at the same 14 monitoring stations, over the same 2022 season (21 June - 16 August). Each photo's filename encoded site, date, and light treatment (e.g. GRTE01_2aug22_w100D); intensity readings that deviated from the bat project's 10/30/50/70/100% schedule were rounded to the nearest level, and repeat IDs of the same site/date/treatment were merged. 9,864 identified specimens across 241 identifications (225 site-nights) were retained, spanning 11 families and 31 subfamilies, dominated numerically by Noctuidae, Lasiocampidae, and Erebididae.

A general activity model -- mirroring the bat Primary Model exactly (color $\times$ intensity + intensity $\times$ dist_km + mean_phase + pct_nonforest + jd_c + jd_c² + (1|site), negative binomial, zero-inflated) -- was fit to total nightly moth counts.

```

Family:      nbinom2  ( log )
Formula:     detections ~ color * intensity + intensity * dist_km +
              mean_phase + pct_nonforest + jd_c + I(jd_c^2) + (1 | site)
Zero infl:   ~1
Data:        insect_env

```

AIC	BIC	logLik	-2*log(L)	df.resid
1992.0	2067.2	-974.0	1948.0	204

Random effects:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	0.0122	0.1103

Number of obs: 226, groups: site, 14

Dispersion parameter for nbinom2 family (): 2.43

Conditional model:

	Estimate	Std.Err	z value	Pr(> z)	
(Intercept)	4.1115	0.2755	14.924	<2e-16	***
colorW	0.0596	0.2689	0.222	0.8244	
intensity30	0.3922	0.2996	1.309	0.1905	
intensity50	0.0481	0.2987	0.161	0.8722	
intensity70	0.1013	0.2827	0.358	0.7201	
intensity100	-0.6289	0.2818	-2.232	0.0256	*
dist_km	0.1086	0.0408	2.661	0.0078	**
mean_phase	-0.4558	0.2414	-1.888	0.0590	.
pct_nonforest	-0.2558	0.2164	-1.182	0.2371	
jd_c	0.0056	0.0036	1.528	0.1266	
I(jd_c^2)	-0.0028	0.0003	-9.562	<2e-16	***
colorW:intensity30	-0.3184	0.3806	-0.837	0.4028	
colorW:intensity50	-0.0686	0.3578	-0.192	0.8479	
colorW:intensity70	0.1072	0.3414	0.314	0.7535	
colorW:intensity100	1.0669	0.3252	3.280	0.0010	***
intensity30:dist_km	0.0490	0.0579	0.846	0.3978	
intensity50:dist_km	0.0393	0.0547	0.718	0.4727	
intensity70:dist_km	0.0264	0.0544	0.484	0.6282	
intensity100:dist_km	-0.1158	0.0523	-2.214	0.0268	*

Unlike bats, moths show no significant color preference at low-to-mid intensity.

At 100% intensity, however, white light produced 3.1× more detections than red ($p < 0.0001$) -- the mirror image of the bat pattern at that same intensity (Section 2), where red and white were statistically indistinguishable. The positive distance slope seen at 10-70% intensity flattens to essentially zero at 100%, echoing the bat finding that the distance gradient weakens at high intensity (Supplemental Figs. S14-S15).

Family-Level Models

Six of nine families cleared the same 30-detection / 20-row threshold used for the bat species models (Noctuidae, Lasiocampidae, Erebidae, Geometridae, Sphingidae, Notodontidae); Crambidae, Drepanidae, and Euteliidae had too few identified specimens to model individually. Noctuidae is numerically dominant (predicted 40-50 detections/night vs. <15 for every other family) and drives most of the aggregate pattern above. At 100% intensity, Erebidae, Noctuidae, and Geometridae each show a 2-3× white-over-red ratio; at lower intensities the families track each other closely with wide, overlapping confidence intervals (Supplemental Figs. S16-S17).

Community Analysis

=== PERMANOVA (adonis2), family-level community ===

	Df	SumOfSqs	R2	F	Pr(>F)	
Model	9	6.781	0.305	5.943	<0.001	***
Residual	122	15.468	0.695			
Total	131	22.249	1.000			

=== dbRDA marginal tests ===

	Df	SumOfSqs	F	Pr(>F)	
color	1	0.201	1.581	0.159	
intensity	4	3.930	7.749	<0.001	***
dist_km	1	1.065	8.401	<0.001	***
mean_phase	1	0.761	6.005	0.001	**
pct_nonforest	1	0.104	0.821	0.494	
brightness_dark	1	0.086	0.682	0.596	
Residual	122	15.468			

=== mvabund --- multivariate GLM (nBoot = 9999) ===

	Dev	Df	Pr(>Dev)
color	6.11	1	0.585

intensity	173.32	4	<2e-16	***
dist_km	73.77	1	<2e-16	***
mean_phase	40.72	1	<2e-16	***
pct_nonforest	13.34	1	0.149	
brightness_d	6.82	1	0.500	

As with bats, light *color* has no detectable effect on moth community composition (dbRDA $p = 0.159$, mvabund $p = 0.585$). Intensity, distance, and moon phase all structure the community strongly (dbRDA $p < 0.001$ for all three; mvabund $p < 2e-16$). Notably, the treatment variables jointly explain far more of the moth community's variation (model $R^2 = 0.305$) than the bat community's (model $R^2 = 0.156$, Community Analysis section above) -- moth community turnover along the light gradient is proportionally stronger than bats'. Per-family univariate tests attribute most of the intensity signal to Lasiocampidae and Noctuidae (both $p < 2e-16$) and the distance signal primarily to Lasiocampidae ($p < 2e-16$) and Noctuidae ($p = 0.005$) (Supplemental Figs. S18-S19).

BAT VS. MOTH COMPARISON

Raw totals differ by roughly an order of magnitude between taxa and reflect different sampling effort rather than treatment response (Supplemental Figs. S20-S22). The figures below instead rescale each model's marginal predictions to % change from that taxon's own baseline, comparing the *shape* of each taxon's response independent of its absolute abundance.

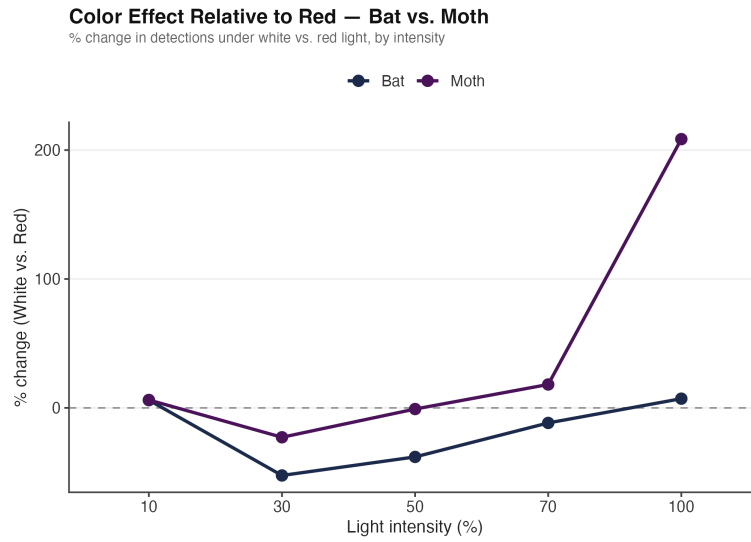


Figure 4: % change in detections, white vs. red light, by intensity -- both taxa on a shared scale.

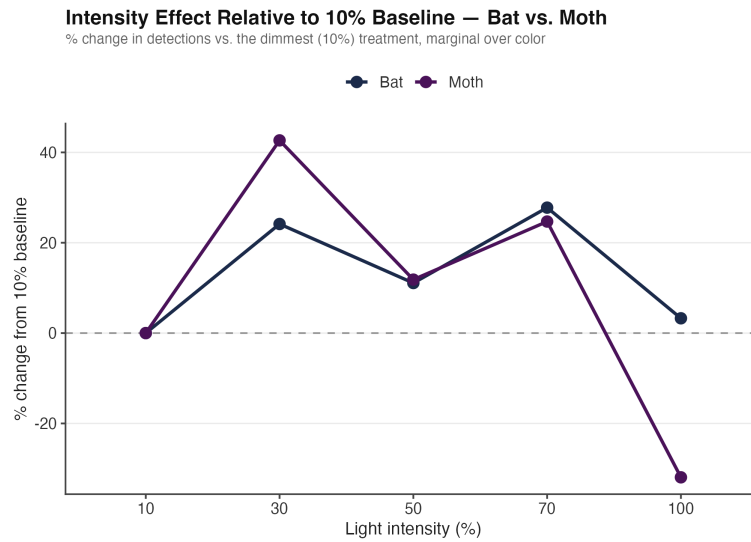


Figure 5: % change in detections from the 10% baseline, marginal over color -- both taxa on a shared scale.

Distance Gradient Relative to the Nearest Shared Site — Bat vs. Moth
 % change in detections vs. 0 km - marginal over color and intensity

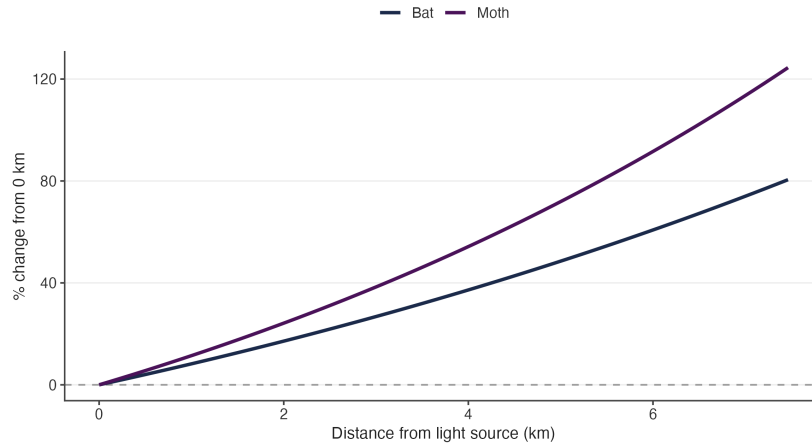


Figure 6: % change in detections along the distance gradient from the nearest shared site -- both taxa on a shared scale.

A companion effect-size forest plot (rate ratios $\pm$ 95% CI for color, each intensity level vs. 10%, and the marginal distance slope) is provided in Supplemental Fig. S23; the same estimates are summarized in the table below.

Term	Bat ratio [95% CI]		Moth ratio [95% CI]	
Color (White vs Red)	0.78	[0.71, 0.86]	1.24	[1.02, 1.51]
Intensity 30% vs 10%	1.24	[0.99, 1.55]	1.43	[0.96, 2.11]
Intensity 50% vs 10%	1.11	[0.86, 1.44]	1.12	[0.71, 1.77]
Intensity 70% vs 10%	1.28	[1.11, 1.47]	1.25	[0.92, 1.69]
Intensity 100% vs 10%	1.03	[0.87, 1.22]	0.68	[0.48, 0.97]
Distance (per km)	1.08	[0.95, 1.23]	1.11	[1.07, 1.16]

The two taxa diverge most clearly on light color: averaged across intensity, bats show 22% fewer detections under white than red (CI excludes 1), while moths show 24% more (CI excludes 1) -- opposite directions. At 100% intensity specifically, bat activity is indistinguishable from the 10% baseline (ratio 1.03, ns) while moth activity drops significantly (ratio 0.68, CI excludes 1): bats hold steady at maximum brightness where moths pull back. Both taxa increase with distance from the Colter Bay parking lot over the shared 0–7.4 km range, and the relative increase is if anything larger for moths (+125% at the far end vs. +80% for bats); only the moth slope is significant on its own (CI excludes 1), because the bat's marginal, intensity-averaged slope is diluted by the reversal at 70–100% intensity documented in the Primary Model (Section 2) -- a reversal not present in the moth data.

SUMMARY

1. Color × intensity interaction: red light at 30–50% produced 1.6–2.1× more bat detections than white light. At 70% and 100%, colors converge. White light is flat across all intensities.
2. Intensity × distance interaction: 70% and 100% intensities show significant negative distance slopes (intensity70:dist_km $\beta = -0.128$, $p < 0.001$; intensity100 $\beta = -0.090$, $p = 0.001$), meaning bat detections under high light intensity increase with distance from Colter Bay parking lot. 30% and 50% slopes are non-significant -- low light intensities are distance-invariant.
3. Community composition is driven by distance, habitat (pct_nonforest), and sky brightness, not by light color.
4. Wing morphology (ear height, forearm length, wing loading, aspect ratio): smaller, slower, more maneuverable species consistently show higher and more distance-sensitive activity than larger, faster, open-air fliers, across every intensity level.
5. Moths mirror bats on intensity and distance (both show a positive distance effect that weakens at 100% intensity, and light color has no effect on community composition in either taxon) but invert the color effect: bats favor red overall while moths favor white, most sharply at 100% intensity (bat ratio 1.03 ns vs. moth ratio 0.68, CI excludes 1).
6. Red light at 30–50% intensity minimises suppression of bat activity while also avoiding the white-light attraction effect seen in moths at high intensity; siting lights >0.5–1.0 km from sensitive habitat remains important for both taxa at high intensity.

SPECIES KEY

Epfu	<i>Eptesicus fuscus</i>	(big brown bat)
Laci	<i>Lasiurus cinereus</i>	(hoary bat)
Lano	<i>Lasionycteris noctivagans</i>	(silver-haired bat)
Myci	<i>Myotis ciliolabrum</i>	(western small-footed myotis)
Myev	<i>Myotis evotis</i>	(long-eared myotis)
Mylu	<i>Myotis lucifugus</i>	(little brown myotis)
Myvo	<i>Myotis volans</i>	(long-legged myotis)

Myyu Myotis yumanensis (Yuma myotis)

Moth families modeled individually (Section 9):

Noctuidae	Owlet moths	(numerically dominant)
Lasiocampidae	Lappet/tent moths	
Erebidae	Erebid moths	(incl. tiger, tussock moths)
Geometridae	Geometer moths	("inchworm" moths)
Sphingidae	Sphinx/hawk moths	
Notodontidae	Prominent moths	

SUPPLEMENTAL MATERIALS

Additional figures supporting the results in the main text, in the order those results are discussed above.

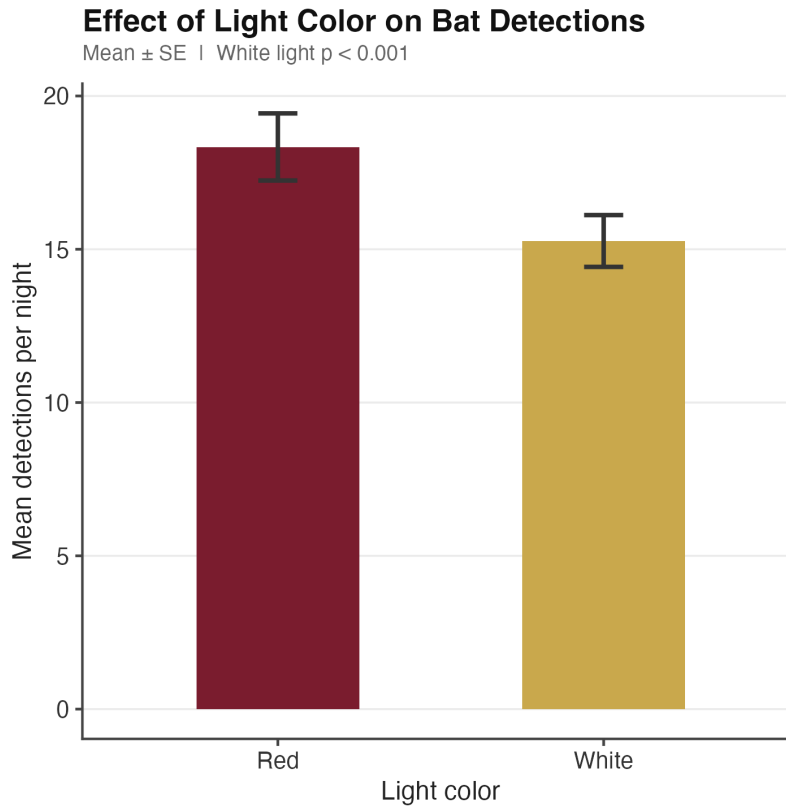


Figure S1: Mean detections per night by light color (cardinal crimson = red, gilded gold = white), averaged across intensities. Error bars = SE.

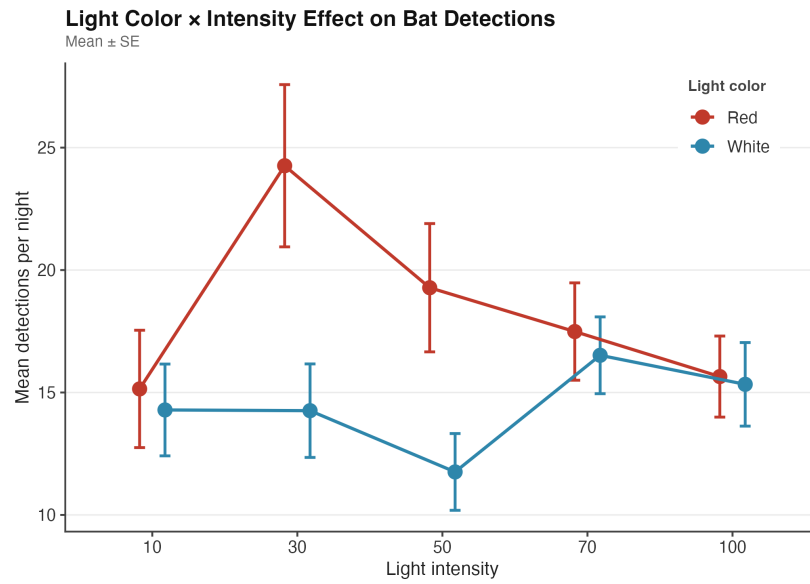


Figure S2: Mean detections per night by color at each intensity level with 95% CI. Cardinal crimson (red) peaks at 30% and converges with gilded gold (white) at high intensity.

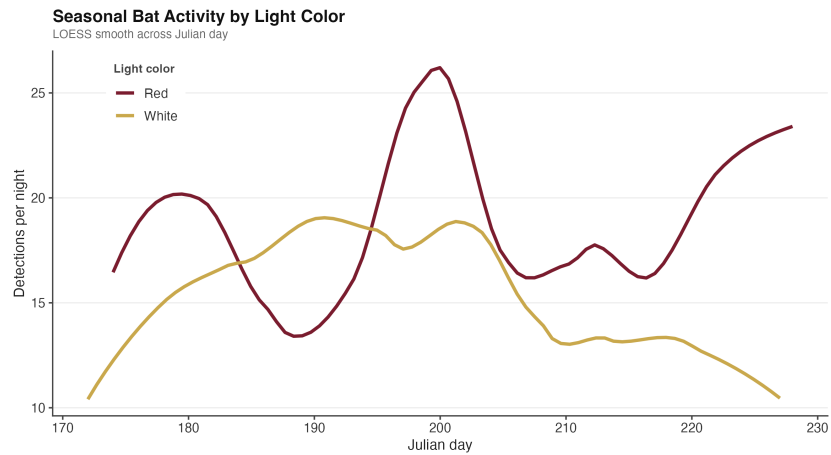


Figure S3: Seasonal trend in bat detections by light color. LOESS smoothed. Cardinal crimson (red) and gilded gold (white) both decline through the season; red/white separation persists across the survey period.

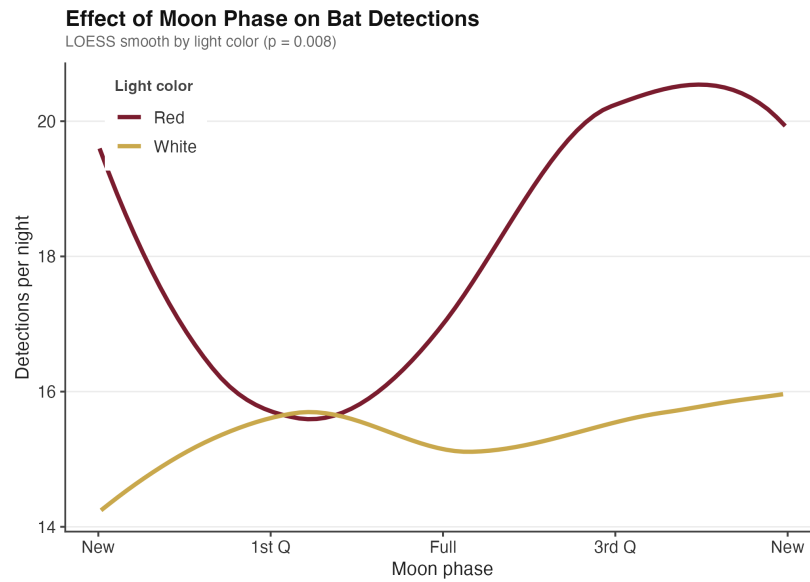


Figure S4: Bat detections vs. moon phase by color (cardinal crimson = red, gilded gold = white). No evident color $\times$ moon interaction.

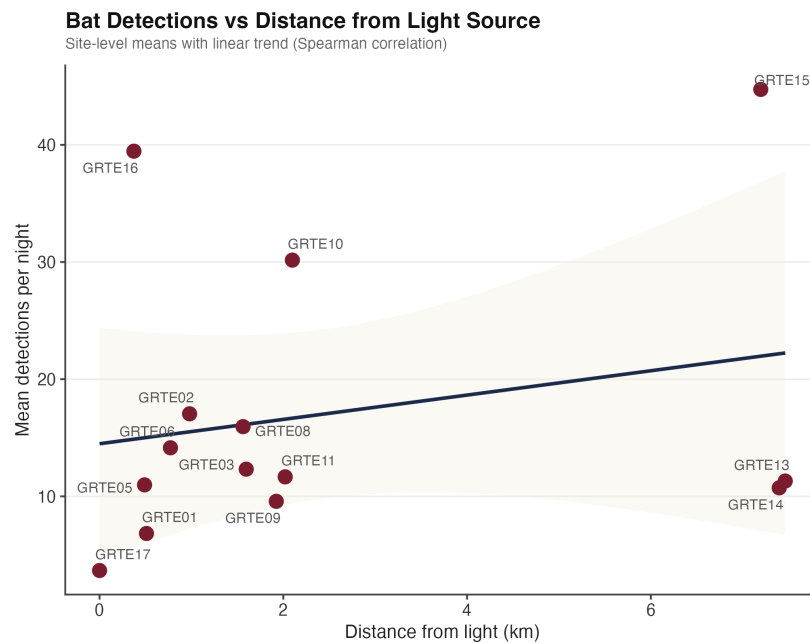


Figure S5: Site-level mean detections vs. distance from Colter Bay parking lot (km). Points in cardinal crimson; navy trend line with manuscript ivory confidence band.

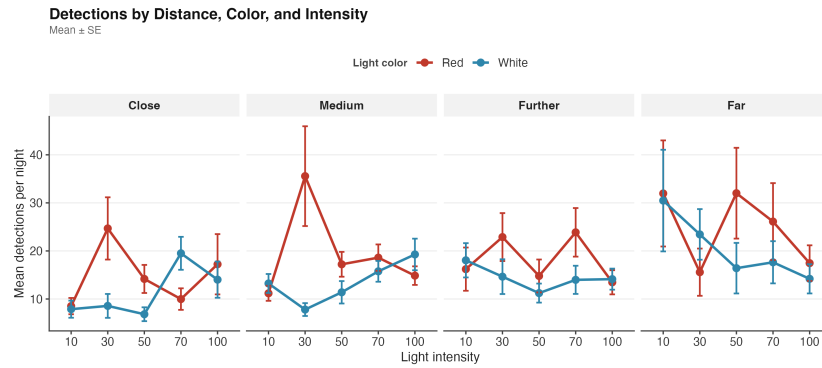


Figure S6: Detections by distance category and light treatment.

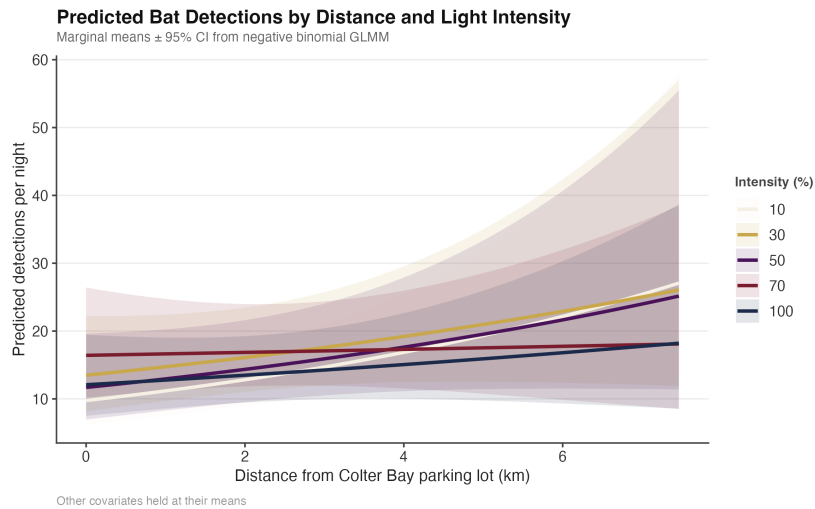


Figure S7: Predicted bat detections across the distance gradient by light intensity level. Marginal means $\pm$ 95% CI from the negative binomial GLMM. Intensity palette: manuscript ivory (10%), gilded gold (30%), papal purple (50%), cardinal crimson (70%), nave navy (100%). Steeper recovery with distance at 70% and 100% reflects the significant intensity $\times$ distance interaction.

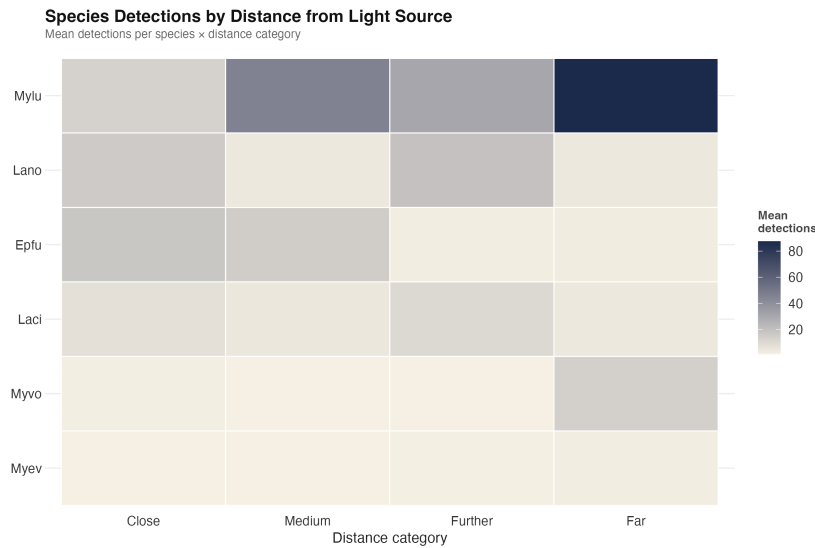
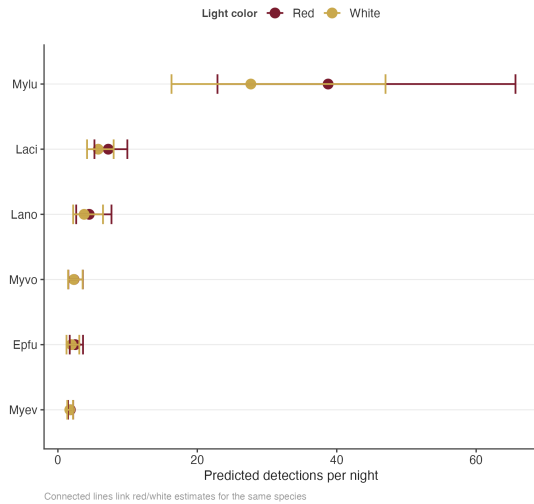


Figure S8: Species activity by distance category (z-scored within species). Scale runs manuscript ivory (low) to navae navy (high).

Species-Level Light Effects — Distance Project

Light Color Effect on Bat Detections by Species

Marginal means $\pm$ 95% CI from per-species negative binomial GLMMs



White vs. Red Light — Detection Ratio by Species & Intensity

Ratio > 1 = more detections under white; < 1 = more under red

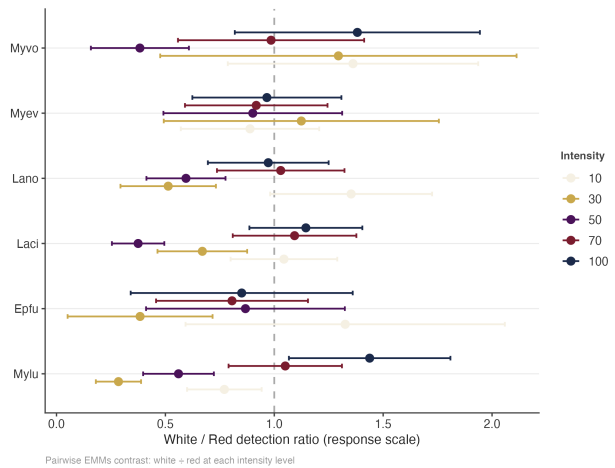


Figure S9: Left: marginal means $\pm$ 95% CI by color (cardinal crimson = red, gilded gold = white), ranked by species. Right: white/red detection ratio by intensity level per species (Catholic intensity palette). Ratio > 1 = more detections under white; < 1 = more under red.

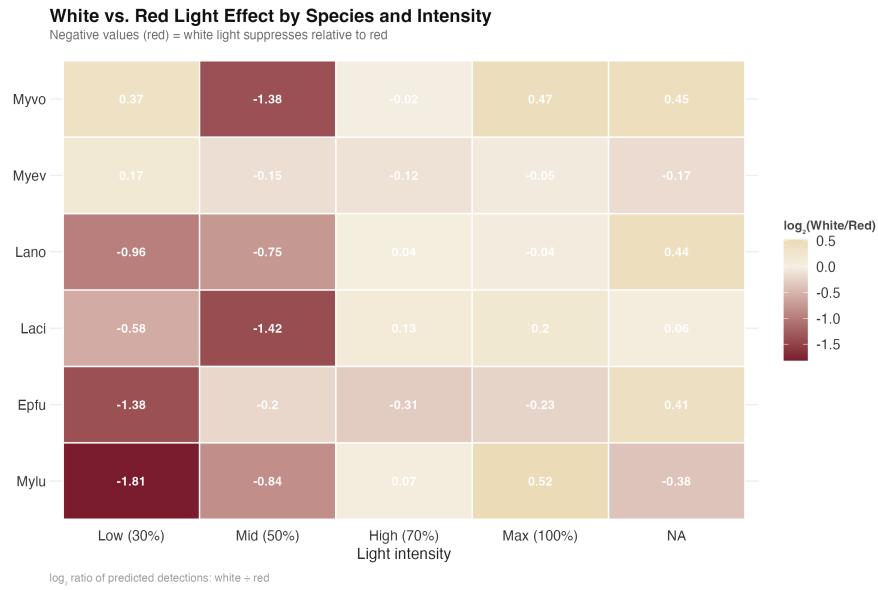


Figure S10: Log₂(white/red) heatmap by species and intensity. Negative values (cardinal crimson) = white suppresses relative to red; positive values (gilded gold) = white enhances. Manuscript ivory = parity.

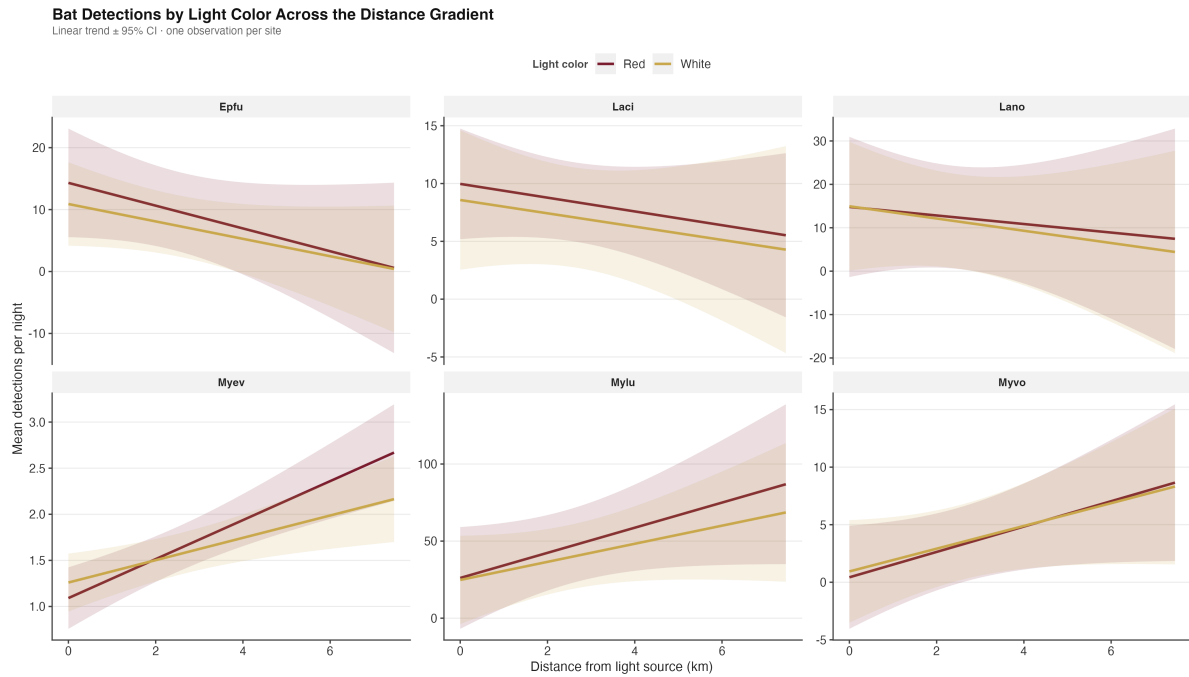


Figure S11: Bat detections by light color across the distance gradient, faceted by species. Linear trend ± 95% CI; one observation per site. Cardinal crimson = red, gilded gold = white.

Bat Community Response by Light Color

Mean detections per species

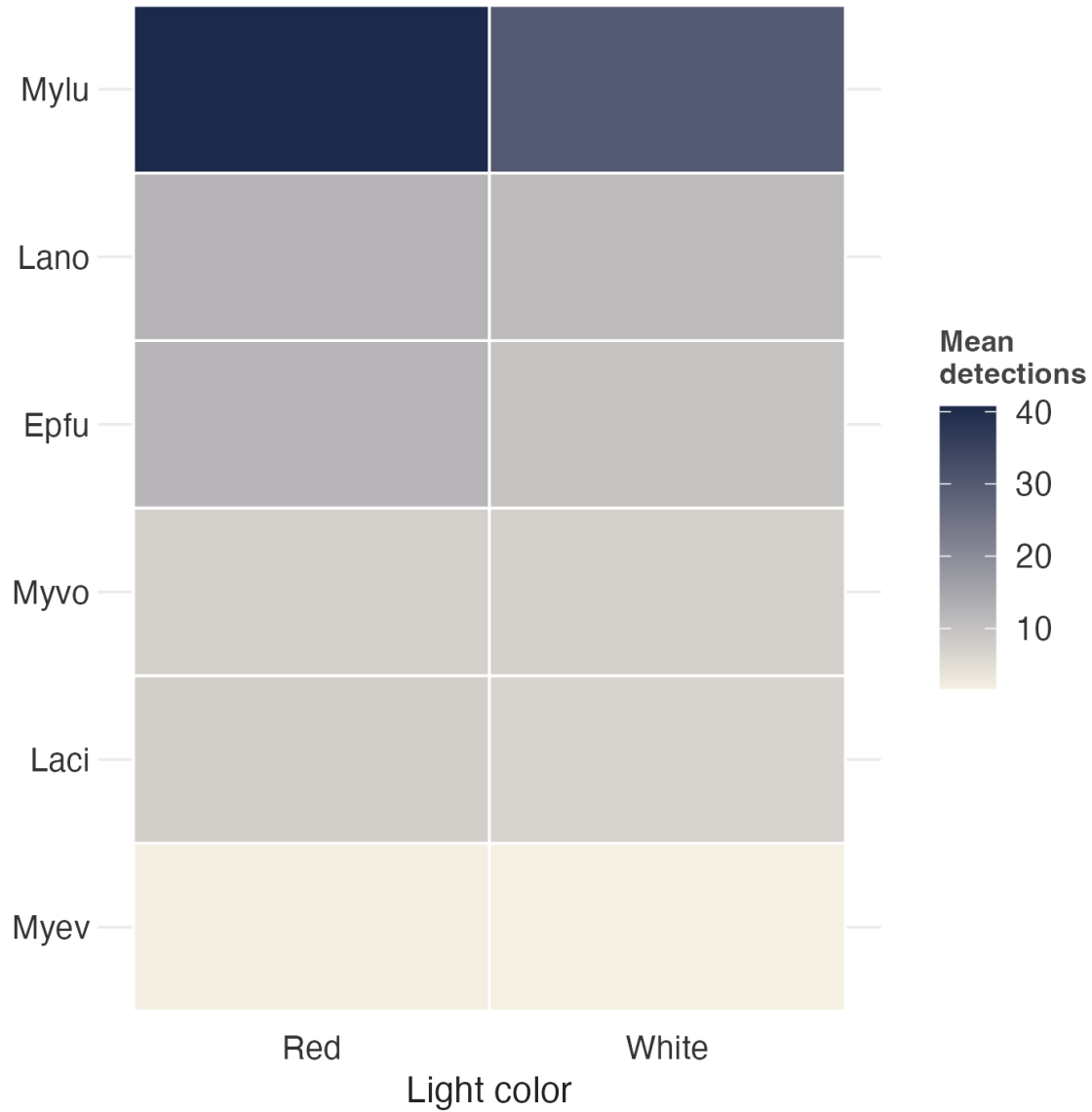


Figure S12: Species activity heatmap (z-scored within species) by light color and intensity. Scale runs manuscript ivory (low) to nava navy (high).

Light Intensity Effect Across Morphology Trait Values

Marginal means $\pm$ 95% CI at low/mean/high (± 1 SD) trait values

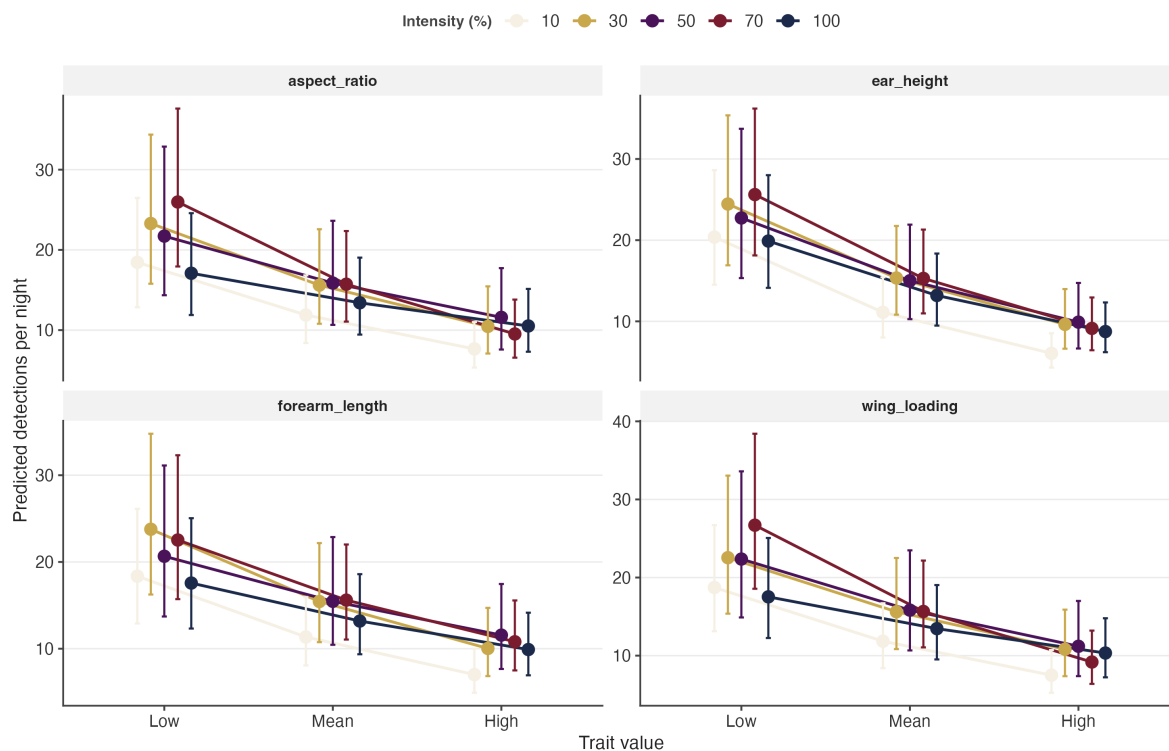


Figure S13: Predicted bat detections by light intensity, evaluated at low/mean/high (± 1 SD) values of each morphology trait.

Bat Activity Across Distance Gradient by Morphology Trait Value

Marginal predictions $\pm$ 95% CI - low/mean/high (± 1 SD) trait values

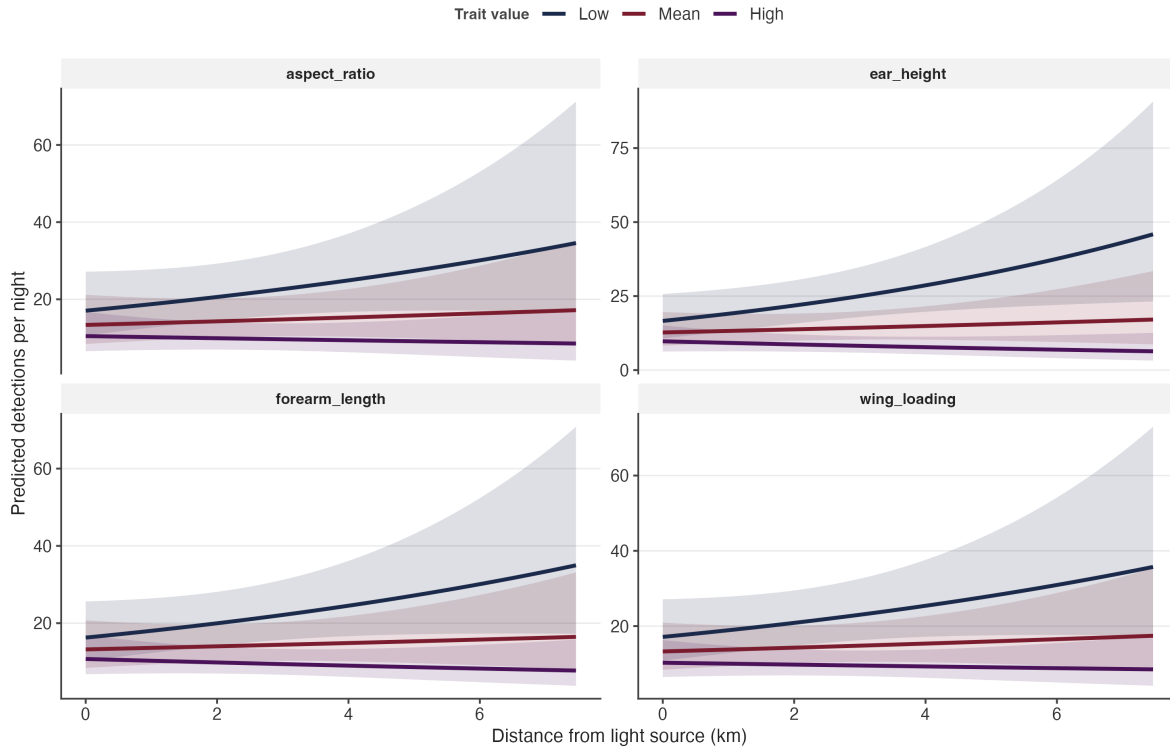


Figure S14: Predicted bat detections across the distance gradient, evaluated at low/mean/high (± 1 SD) values of each morphology trait.

Light Color $\times$ Intensity Effect on Moth Detections

Marginal means $\pm$ 95% CI from negative binomial GLMM

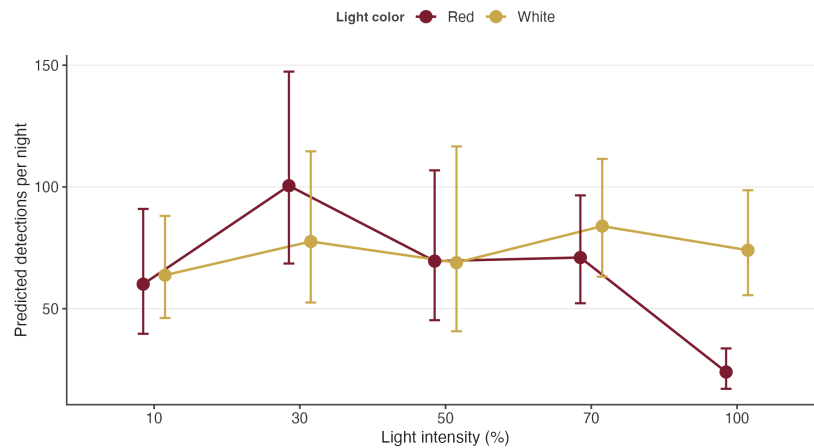


Figure S15: Moth detections by light color and intensity (general activity model). Cardinal crimson = red, gilded gold = white.

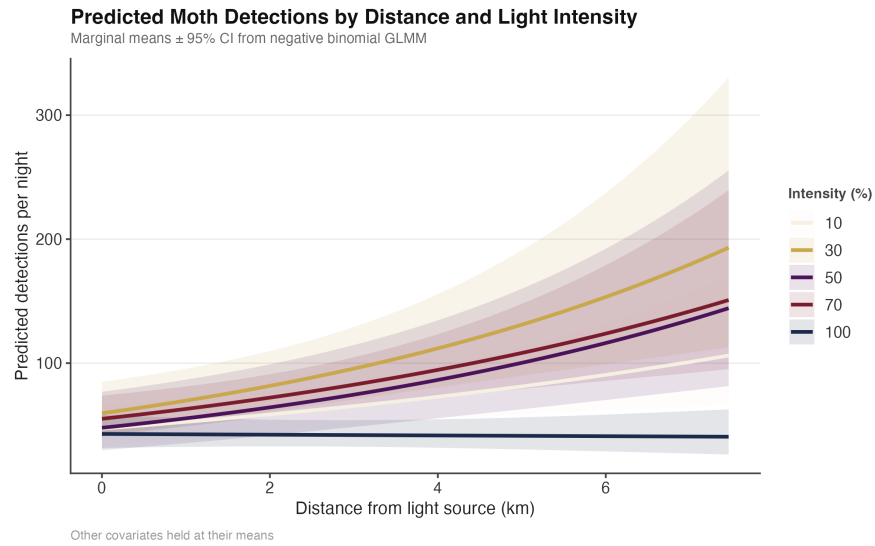


Figure S16: Predicted moth detections by distance and light intensity.

Family-Level Light Effects — Distance Project

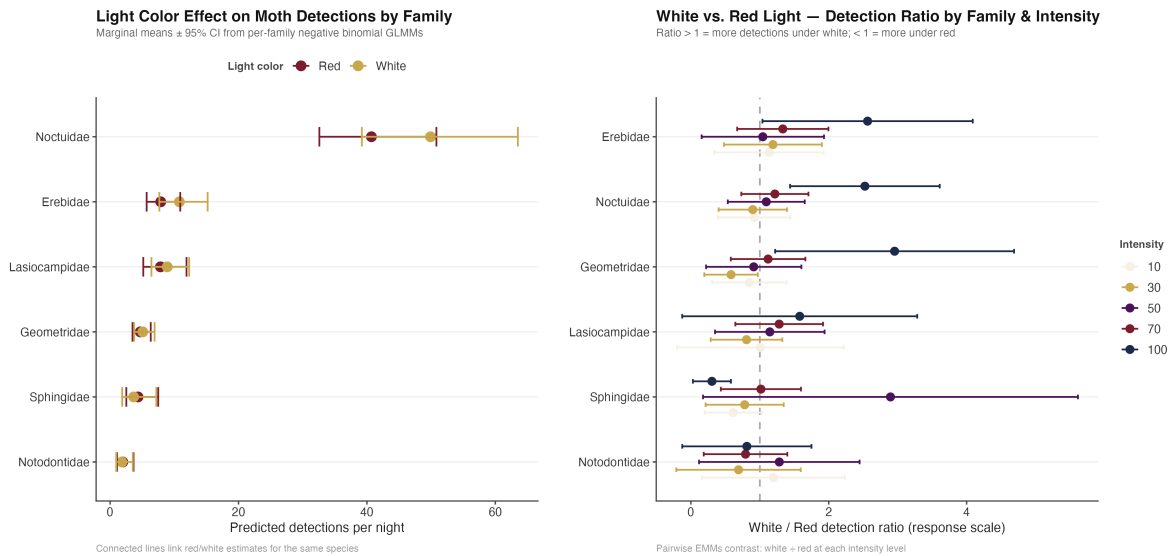


Figure S17: Left: marginal means $\pm$ 95% CI by color, ranked by family. Right: white/red detection ratio by intensity level per family. Ratio > 1 = more detections under white; < 1 = more under red.

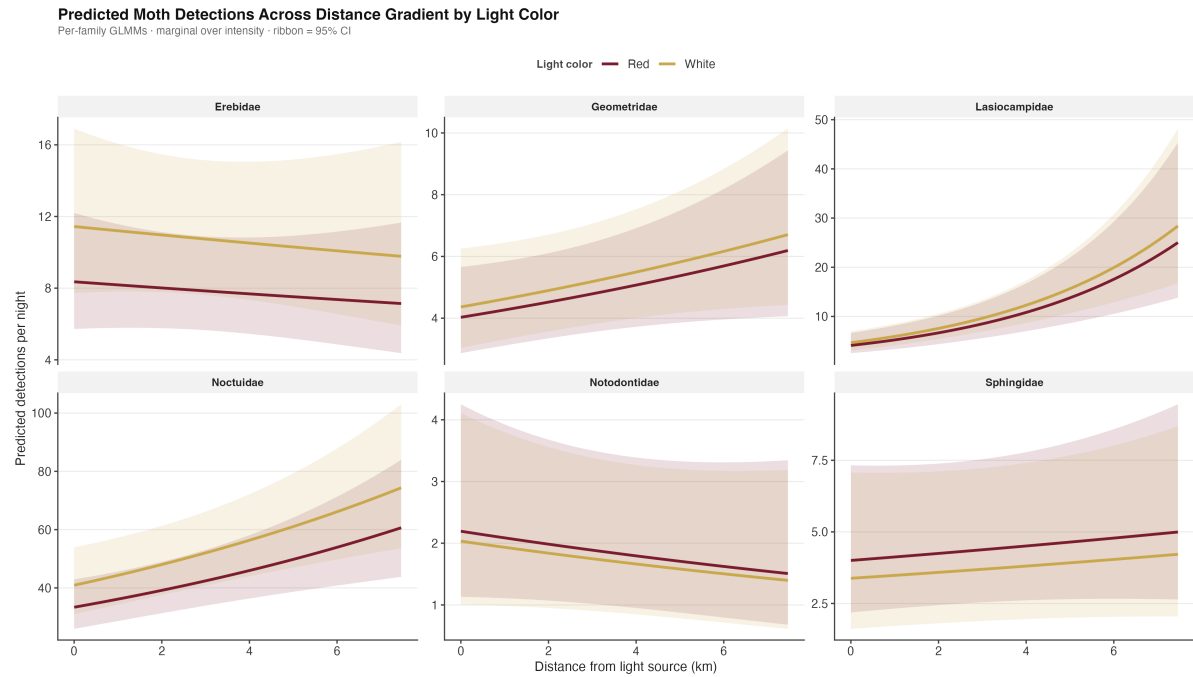


Figure S18: Predicted moth detections across the distance gradient by light color, faceted by family.

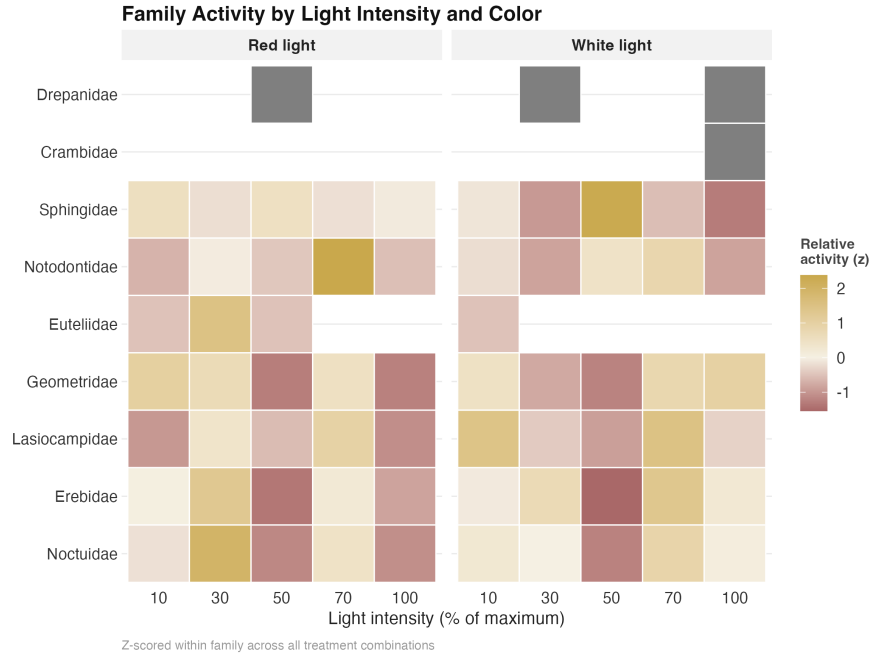


Figure S19: Moth family activity by light intensity and color (z-scored within family).

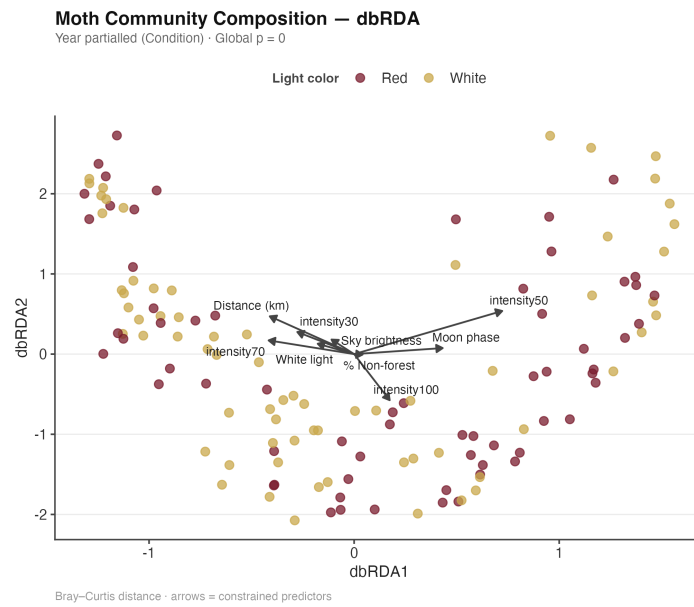


Figure S20: Moth family community composition -- dbRDA ordination.

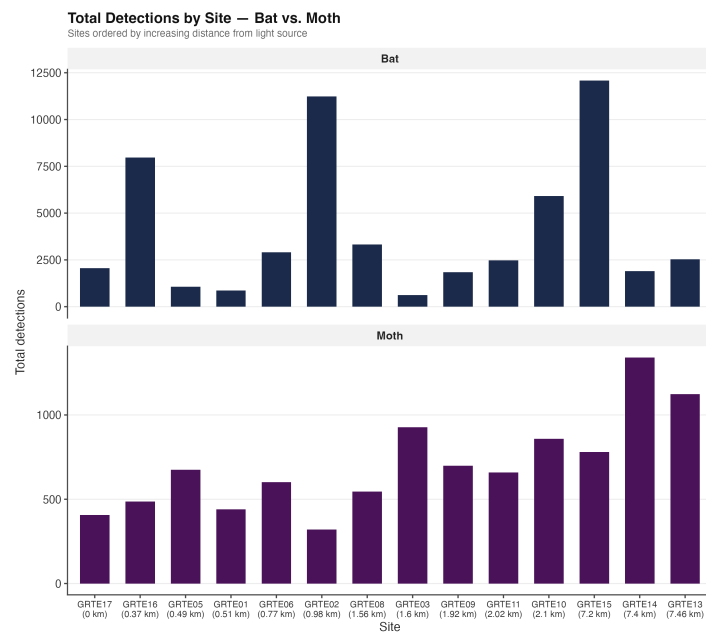


Figure S21: Total detections by site, bat vs. moth (sites ordered by distance from the Colter Bay parking lot). Bat = nave navy, Moth = papal purple.

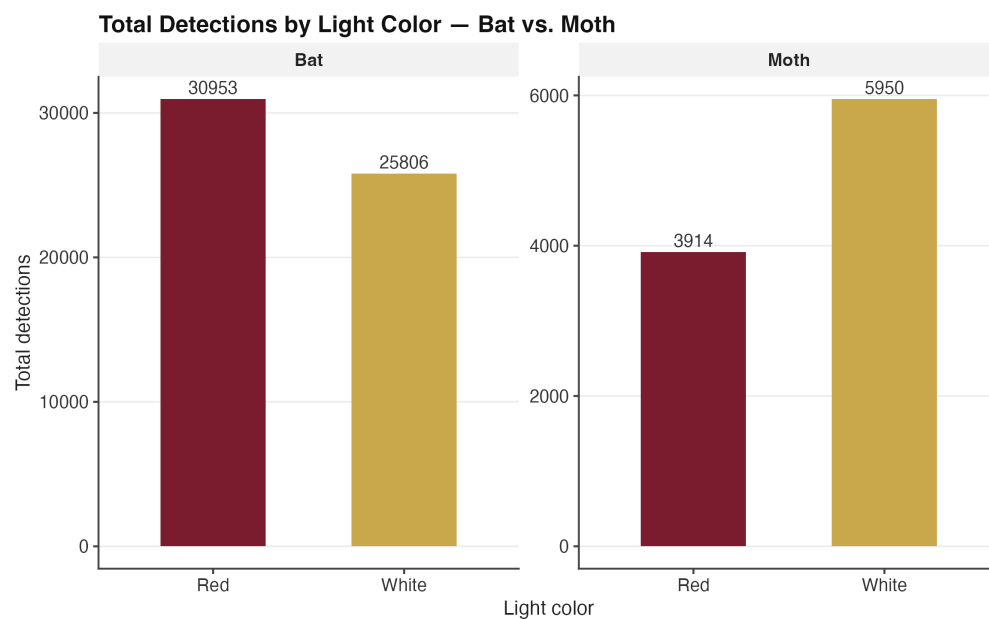


Figure S22: Total detections by light color, bat vs. moth.

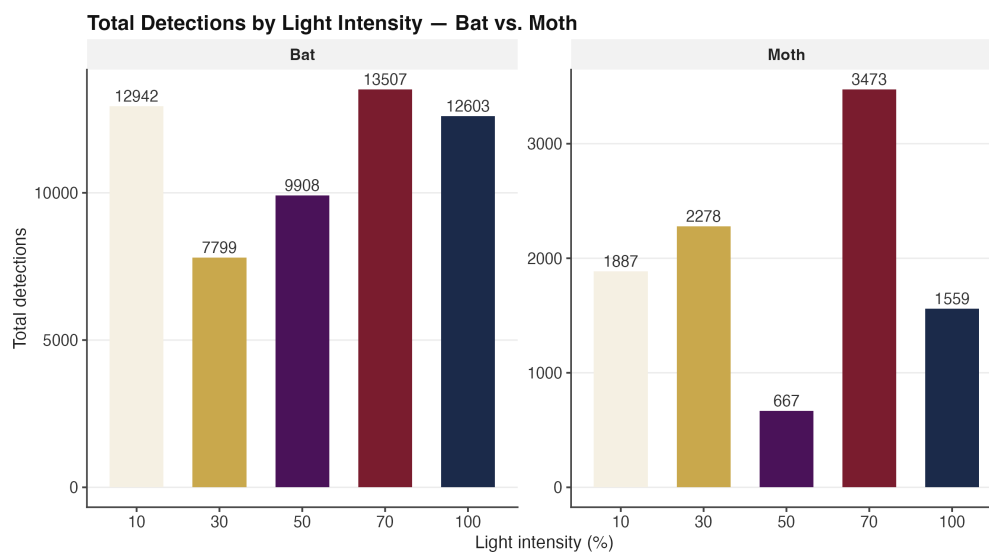


Figure S23: Total detections by light intensity, bat vs. moth.

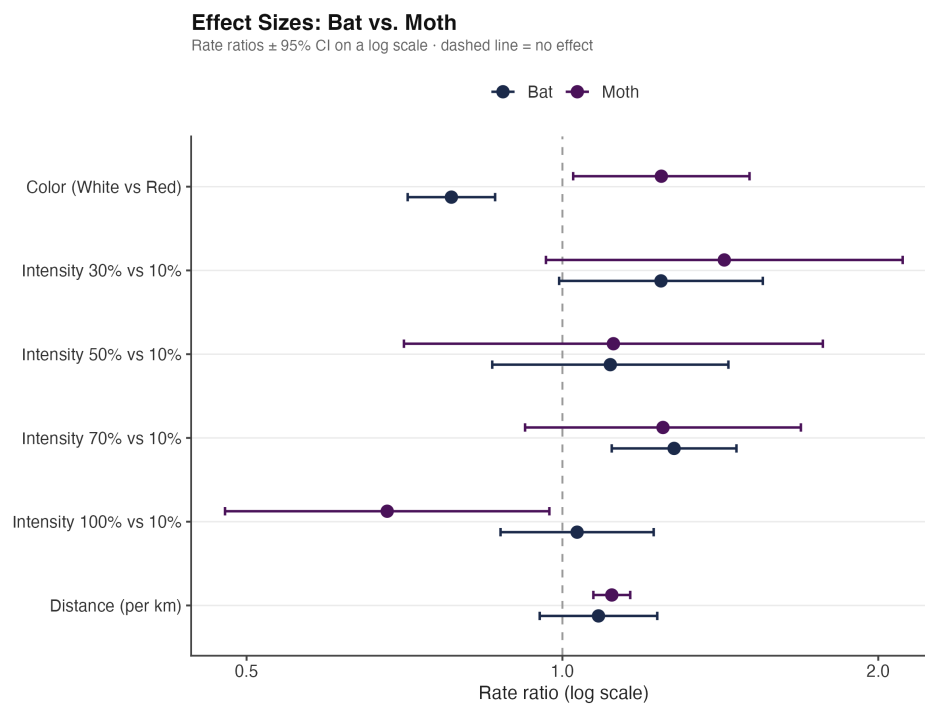


Figure S24: Effect-size comparison: rate ratios $\pm$ 95% CI for color, each intensity level vs. 10%, and the marginal distance slope.